

VCE Mathematical Methods (MM34) — Term 1 Examination 2

AI-Marked Submission · Marker: Trainer (AI Assistant) · 2026-05-26

29 / 40**Overall: 72.5% — Normal.**

Strong across Q1 (perfect) and Q2 (8 / 10). Focus areas: trig stationary points (Q3c) and careful derivative substitution (Q3f). The multiple-choice section was 5 / 10 — please verify the answer key and review each incorrect item.

Section	Score	%	Key Notes
Section A — MC	5 / 10	50%	Q2, 4, 6, 8, 9 wrong — verify assessment key
Q1 — Rectangle & Parabola	10 / 10	100%	All steps correct — excellent work
Q2 — Transformations	8 / 10	80%	Q2d interval direction wrong
Q3 — Trig Functions	6 / 10	60%	Q3c SP analysis, Q3f derivative-argument errors
TOTAL	29 / 40	72.5%	Strong Q1, good Q2, need work on trig SPs

Section A — Multiple Choice (5 / 10)

Score: 5 / 10. Each question is worth 1 mark; no method marks.**Correct:**

Q1: C Q3: A Q5: A Q7: D Q10: B

Incorrect:

Q#	Given	Correct	Note
Q2	D	C	Answer should be C — possible misreading of the question.
Q4	B	A	Answer should be A — re-check the working.
Q6	C	D	Answer should be D — check domain restrictions.
Q8	B	D	Answer should be D — review the concept tested.
Q9	C	D	Answer should be D — check the final simplification step.

Note: a previous, similar multiple-choice section scored 9 / 10. Verify the correct answer key independently to rule out an answer-key discrepancy.

Q1 — Rectangle & Parabola (10 / 10)

Result: FULL MARKS — excellent work across all parts.**Q1a(i) [1]** $A(t) = t(4 - t^2)$ correct expression**Q1a(ii) [2]** $t = 2 \cdot \sqrt{3} / 3$, $A_{\max} = 16 \cdot \sqrt{3} / 9$ both correct**Q1b(i) [1]** $y = -2px + 4 + p^2$ correct tangent**Q1b(ii) [1]** $S = (0, 4 + p^2)$, $U = ((4 + p^2) / 2p, 0)$ both coordinates correct**Q1b(iii) [1]** $\text{Area} = (4 + p^2)^2 / 4p$ correct area expression**Q1b(iv) [2]** $p = 2 \cdot \sqrt{3} / 3$, $\text{min area} = 32 \cdot \sqrt{3} / 9$ correct**Q1c [2]** $p = \sqrt{3k} / 3$, $\text{min area} = 4k \cdot \sqrt{3k} / 9$ correct general solution**Q1 Total: 10 / 10** — outstanding. All working clear and algebra precise.

Q2 — Transformations (8 / 10)

Question: $f(x) = -x^2 + 4x$. Transformations: dilation by factor $2/a$ from the y-axis, dilation by $1/a$ from the x-axis, translation c units up.

Q2a [2/2] $g(x) = -2x^2/a^2 + 4x/a + c$, domain $[0, 2a]$ correct expression and domain

Q2b [1/1] vertex $(a, 2 + c)$ correct

Q2c [2/2] intersections at $x = 0$ and $x = 2a/(a + 1)$ correct

Q2d [0/2] correct answer: $x = 0$ together with $[3/2, 2]$ wrong interval direction

The submitted answer $[0, 3/2]$ is the opposite interval direction: the condition requires values at or beyond $3/2$, not between 0 and $3/2$.

Q2e [3/3] $x = a/(a + 1)$, $c = 2(a - 1)/(a + 1)$ correct working and condition

Q2 Total: 8 / 10 — strong algebraic work; revisit interpreting interval conditions (Q2d).

Q3 — Trig Functions $f(x) = 3\sin(2x) + \sin(6x)$ (6 / 10)

Q3a [1/1] period = π correct

Q3b [2/2] $f(x) > 0$ on $(n\pi, n\pi + \pi/2)$ correct domain

Q3c [1/3] $f'(x) = 6\cos(2x) + 6\cos(6x) = 0$ derivative set up correctly

Errors in the stationary points: found $x = \pi/6$ (should be $\pi/8$) and $5\pi/6$ (should be $5\pi/8$); the y-value $\sqrt{2}$ at $x = \pi/4$ should be $2\sqrt{2}$. Only $x = \pi/4$ and $3\pi/8$ were correct. Enumerate the $\cos(4x)\cos(2x) = 0$ branches systematically.

Q3d [1/1] range $[-2\sqrt{2}, 2\sqrt{2}]$ correct

Q3f [0/2] solve $f'(\pi/9) = f'(2\pi/9)$ incorrect derivative evaluation

The wrong derivative arguments were substituted; the final answer of 59 degrees differs from the solution. Substitute carefully and check each term before combining.

Q3g [1/1] 6 distinct values correct count

Q3 Total: 6 / 10 — Q3a, b, d, g done well; Q3c and Q3f need attention.

Final Summary

Total Score: 29 / 40 (72.5%) — Normal

Strengths

- Q1: perfect 10 / 10 — excellent algebraic manipulation and optimisation.
- Q2: strong on transformations (8 / 10) — almost all parts correct.
- Q3a, Q3b, Q3d: good grasp of trig period, sign analysis, and range.
- Q3g: correct count of distinct values.
- Working is generally well-structured and clearly laid out.

Areas for improvement

- Q2d: interval conditions from inequality analysis — mind the direction of the intersection sets.
- Q3c: solving $\cos(4x)\cos(2x) = 0$ — enumerate solutions from each factor systematically.
- Q3c: verify stationary-point y-values by substituting back into $f(x)$.
- Q3f: substitute carefully into the derivative — check each term before combining.
- Multiple choice: 5 / 10 is below the usual standard — verify the answer key and review each incorrect item.